

Question Number	Answer	Additional guidance	Mark
4(a)(i)	An explanation that includes three of the following points: - oxygen for (aerobic) respiration (1) - glucose for {respiration / ATP production} / amino acids for protein synthesis (1) - optimum temperature for {enzyme / metabolic} reaction (1) - water for {hydrolysis reactions / solvent} (1) - optimum pH for {enzyme / metabolic} reaction (1)	accept lack of oxygen for survival of obligate anaerobes accept lipids for synthesis of cell membranes accept suitable stated temperature for {faster/optimum} enzyme activity accept to prevent dehydration accept suitable stated pH for {faster/optimum} enzyme rate of reaction	(3)

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	An explanation that makes reference to two of the following points: - oil-based plastic is non-renewable (1) - oil-based plastic {takes a long time to decompose / isn't biodegradable} (1) - contributes to increased carbon dioxide (in atmosphere) / not carbon neutral (1)	accept (oil is a) finite resource / may run out / may not always be available for future generations accept contributes to {global warming/ greenhouse effect}	(2)

Question Number	Answer	Additional guidance	Mark
4(b)(i)	An answer that makes reference to three of the following: • bacterial survival decreases over time (on all three boards) (1) • bacterial survival is lowest on pine chopping board (1) • no change in number of bacteria surviving on plastic during first 4 hours (1) • relevant comment about validity of data (1)	accept converse for plastic accept only significant difference between spruce and plastic is at 4 hours / only slight difference between spruce and plastic at 24 hours accept same survival for {spruce/plastic} after 1 hour e.g. no reference to repeats, no error bars	(3)

Question Number	Answer	Additional guidance	Mark
4(b)(ii)	An answer that includes one of the following points: • pine boards have antimicrobial properties (1) • pine chopping boards have a pH outside of bacterial optimum pH (1)	accept pine boards contain chemicals which kill bacteria accept converse for plastic ignore temperature and water	(1)